Cities 166 (2025) 106196

Contents lists available at ScienceDirect

Cities

journal homepage: www.elsevier.com/locate/cities

Measuring urban physical environments using image deep features

Yingjing Huang, Fan Zhang [*], Lun Wu, Yu Liu

Institute of Remote Sensing and Geographical Information System, School of Earth and Space Sciences, Peking University, Beijing, China

ARTICLE INFO

Keywords:
Street-level imagery
Urban imagery
Computational representation
Representation learning
Urban landscape
Urban typology

ABSTRACT

Effective and efficient representation of the urban physical environment is essential in urban studies. Street-level imagery, a prevalent data source for studying urban physical environments, provides detailed visual information about cityscapes. However, traditional methods typically rely on the audits of predefined semantic elements, which often oversimplify the urban environments. This study introduces a framework that applies image deep features to capture the complexity of urban physical environments, thereby inspiring solutions to urban problem. These deep features encode rich urban landscape semantics into a high-dimensional feature, capturing not only visual elements but also their detailed attributes and complex spatial interrelationships. We validate this approach through a case study on urban landscape cell identification, a task that demands an integrated understanding of urban visual elements and their spatial organization. By analyzing over 8 million street-level images from 36 Chinese cities, we demonstrate that these deep features effectively encode complex patterns and latent structural characteristics of urban environments, surpassing traditional methods. This work offers a novel perspective by introducing image deep features into urban studies, fundamentally enhancing urban understanding and providing a powerful foundation for future research and informed decision-making.

1. Introduction

The urban physical environment not only reflects how humans shape urban spaces but also acts as a dynamic interface for human-urban interactions, reflecting and recording attributes such as residents' activity patterns and socioeconomic dynamics (Gebru et al., 2017; Naik et al., 2017; Yeh et al., 2020). It involves the physical structure and spatial composition of urban spaces, including specific elements such as buildings, roads, infrastructure, and green spaces. From architectural forms to transportation systems, these physical elements are the fundamental interfaces through which people interact with the city in their daily lives. Numerous studies have discussed that the urban physical environment reflects not only urban functions and the dynamics of human activities but also multiple socioeconomic attributes (Branas et al., 2018; Guo & Loo, 2013; Van Cauwenberg et al., 2012).

In recent years, street-level imagery has become an important and rapidly developing data source for studying the urban physical environment (Biljecki & Ito, 2021; Zhang et al., 2024). Such datasets contain images of urban street scenes that simulate human perspectives and can be accessed through online map services such as Tencent Street View and Google Street View. The high-resolution and ground-level perspectives of street-level imagery greatly enrich descriptions of the urban physical environment. For example, researchers can observe detailed features of building façades through these images, such as materials, colors, decorative styles, and the spatial relationships between buildings (Sun et al., 2022). Capturing such details allows for a more comprehensive and refined understanding of the urban environment, providing robust data support for urban planning and design.

Detecting visual elements in semantic spaces has been an important method for representing the urban physical environment using street-level imagery. Researchers have manually measured elements in street-level images, such as sidewalk facilities and parking spaces, to characterize the urban physical environment, thereby supporting analyses in areas such as resident health and bikeability (Hoedl et al., 2010; Kelly et al., 2013; Rundle et al., 2011). With the advancement of artificial intelligence technologies, current computer vision techniques—such as semantic segmentation and object detection—have been widely applied to the quantitative analysis of street-level imagery, enhancing studies of the urban physical environment. By measuring the number or pixel proportions of visual elements in street-level images, researchers can describe urban visual characteristics from multiple dimensions (Huang et al., 2025; Li et al., 2015; Liu et al., 2019).

However, relying solely on quantifying visual elements in semantic spaces is insufficient to represent their complexity. First, the complex

* Corresponding author at: Peking University, No. 5 Yiheyuan Road, Haidian District, Beijing 100871, China.
E-mail address: fanzhanggis@pku.edu.cn (F. Zhang).

https://doi.org/10.1016/j.cities.2025.106196
Received 11 February 2025; Received in revised form 10 June 2025; Accepted 19 June 2025
Available online 2 July 2025

attributes of visual elements are difficult to fully capture. For example, the study by Gebru et al. (2017) found that by analyzing the brands, models, and years of cars in street-level images, one can accurately estimate socioeconomic characteristics such as income, race, education, and voting patterns. Similarly, vegetation attributes include not only species but also health status. Although existing datasets have categorized visual elements into buildings, sidewalks, and other classes through extensive manual annotation (Zhou et al., 2017, 2019), annotating such complex attributes requires expert knowledge and enormous manual labor, making this goal nearly impossible to achieve (Cordts et al., 2016; Ros et al., 2016).

Secondly, the spatial arrangements and complex relationships among visual elements are also difficult to fully represent. Even if the pixel proportions of certain visual elements are similar, different urban areas may exhibit vastly different spatial relationships and scene characteristics, which are hard to capture with existing methods. For example, street trees neatly arranged along urban roads and woodlands in the suburbs may have similar green view index values, but due to significant differences in spatial layout, the urban physical environment semantics they convey are entirely different. Simple representations of visual elements may lead to inaccurate analyses of the urban physical environment, potentially causing bias in critical applications such as urban planning, resource allocation, and policy-making.

The development of representation learning offers a promising approach for comprehensively expressing the semantic information of the urban physical environment. Compared to methods that rely on expert knowledge for manual feature engineering, representation learning enables data-driven extraction of high-level features directly from raw data, facilitating more nuanced and scalable understanding (Hou et al., 2024; Luo et al., 2023; Mai et al., 2023; Mikolov et al., 2013).

Deep features are high-dimensional vectors extracted from intermediate layers of deep neural networks, including convolutional neural networks (CNNs), Transformer-based architectures, and large-scale foundation models. These features capture complex spatial structures, semantic content, element relationships, and contextual cues within images. Unlike handcrafted features, which require manual design and domain knowledge, deep features are automatically learned from large datasets. As the network depth increases, these features transition from low-level patterns such as edges and textures to high-level semantic representations, including object categories and scene understanding (He et al., 2016; Krizhevsky et al., 2012; Vaswani et al., 2017). In the geospatial domain, representation learning methods such as MT-POI2Vec (Yao et al., 2023) and Place2Vec (Yan et al., 2017) have been successfully applied to extract deep features from spatial data, improving the robustness and accuracy of geographic modeling and supporting multimodal data integration within GeoAI systems. For image-based applications, scene classification models trained on datasets like Places365 not only identify scene categories but also capture latent attributes such as built environment features, human activities, and lighting conditions (Zhou et al., 2018). Although high-dimensional deep features are often difficult to interpret directly, they serve as a powerful foundation for computational models to perform downstream tasks efficiently.

This study introduces a framework that applies image deep features to effectively and efficiently characterize urban physical environments, thereby supporting the solution of urban problems. We validate this approach through a case study focused on urban landscape cell identification. This task is ideal for evaluating deep features' semantic richness and applicability because it demands an integrated understanding of visual elements' attributes and spatial arrangements, requiring a comprehensive representation of the urban physical environment's spatial structure and intricate relationships. This study proposes and validates an effective deep feature-based approach for representing urban landscapes, supported by the construction of a comprehensive large-scale dataset of street-level images that facilitates systematic intercity comparisons. Furthermore, our analysis reveals previously unexplored urban landscape patterns and their associations with socioeconomic factors.

This study presents a novel perspective by incorporating image deep features into urban studies, thereby significantly enhancing our ability to understand, analyze, and ultimately address complex urban problems. By providing a rich, data-driven approach for characterizing physical environments, our study opens new avenues for quantitative urban research. This not only deepens insights into urban form and function but also provides urban planners and policymakers with powerful tools for evidence-based decision-making.

2. Related work

2.1. Images in urban studies

Images have been a critical source of data for studying the urban physical environment. Early researchers typically relied on manually collected photos and sketches to audit and analyze physical characteristics, such as rhythm, color, complexity, and hierarchical element structures across multiple dimensions (Wohlwill, 1976). These early studies primarily focused on the aesthetic values inherent in urban form (Howard, 1902; Miller, 1910) and later evolved to systematically explore human perception and evaluation of urban environments (Arnheim, 1954). However, these pioneering works were often limited by low-resolution and restricted-coverage image datasets, resulting in small sample sizes that constrained the generalizability and representativeness of their findings.

The rapid development of urban sensing technologies has significantly transformed the collection and analysis of urban images. Emerging sources, such as satellite imagery, street-level imagery, drone photography, and sensor networks, have overcome past data limitations, providing high-resolution, extensive urban image data. These advancements enable more detailed analysis of contemporary urban environments, broadening the spatial and temporal dimensions of studies and enhancing their representativeness and generalizability (Zhang et al., 2018). Consequently, researchers can now more comprehensively and accurately capture features and changes in urban environments, meeting the demand for in-depth analyses of contemporary urban phenomena (Dou & Chen, 2017; Rapinel et al., 2014; Sun et al., 2022).

Complementing these data advancements, breakthroughs in artificial intelligence (AI) and computer vision have provided powerful support for the automated analysis of large-scale urban image data. These technologies enable the training of deep learning models to rapidly extract and quantify explicit features of the urban physical environment, such as building footprints and green space coverage, from massive image volumes (Zhang et al., 2017). The effectiveness of these AI-driven analyses heavily relies on how images are represented and processed by computer vision models, which we will discuss in the next section.

2.2. Image representation in computer vision

Image representation is a core issue in the field of computer vision (CV), concerning how computers interpret and understand visual data. Traditional computer vision methods primarily relied on hand-crafted shallow features, such as color histograms, texture descriptors, and shape features, to extract semantic information from images (Long & Liu, 2017). Typical algorithms included Scale-Invariant Feature Transform (SIFT) (Lowe, 1999) and Histogram of Oriented Gradients (HOG) (Dalal & Triggs, 2005). However, these methods were highly dependent on expert knowledge, often struggled to capture complex semantic structures within images, and were inefficient at handling large-scale data.

With the advancements in computational power and data availability, deep learning has revolutionized the field of image representation, and deep features have become widely used and proven effective. Deep learning, through multi-layered neural networks, can automatically learn high-level features in images, eliminating the need for manually designed features and significantly improving representation accuracy and generalizability (Krizhevsky et al., 2012). Currently, deep learning-driven image representations fall into two main categories: supervised learning and self-supervised learning. Supervised learning relies on large amounts of labeled data, training models by minimizing the difference between model predictions and true labels. However, obtaining high-quality, large-scale labeled data is costly, and the deep features extracted through supervised learning are often constrained to shallow representations tied to label information, limiting the model's generalizability (Doersch et al., 2015). Self-supervised learning has emerged as a popular research focus to address these limitations. Self-supervised learning designs pretext tasks based on the structure of the data itself to learn effective feature representations, thereby reducing the reliance on labeled data and supporting the learning of deeper feature representations (Chen et al., 2020; He et al., 2020). Deep features have been widely applied in computer vision tasks such as image retrieval, image classification, and object detection, demonstrating substantial effectiveness.

While street-level imagery serves as a critical medium for understanding urban physical environments, its analysis using computer vision has primarily focused on extracting predefined semantic elements (e.g., greenery, building facades) through techniques like semantic segmentation (Kelly et al., 2013; Rundle et al., 2011). These approaches support tasks such as urban function classification, urban profile prediction, and perception modeling (Fan et al., 2023; Liang et al., 2023). However, relying solely on predefined semantic labels may limit the discovery of more complex or latent visual patterns inherent in street-level imagery. To overcome this and capture richer, high-dimensional characteristics of urban landscapes, our study explores how deep features can represent and compare the spatial structure of urban environments across cities, offering new perspectives for data-driven urban analysis.

3. Deep feature framework for urban analysis

This study proposes a framework to effectively represent and analyze the urban physical environment using deep features (Fig. 1). The framework utilizes high-dimensional deep features extracted from street-level images to support urban tasks and uncover the underlying patterns. It contains four parts. First, the **feature extraction** utilizes a deep learning model to obtain image representations, then aggregates them at specific spatial units based on the downstream task. Second, the **feature understanding** further analyzes deep features to understand their underlying mechanisms. Third, the **feature utilization for tasks** applies these features to solve specific urban problems. Last, the **feature exploration** utilizes deep features to uncover potential underlying urban patterns.

To demonstrate the applicability of the proposed framework, we conducted a case study on urban landscape cell identification. Guided by our four-part framework, we extracted and grid-aggregated deep features from street-level images. We then interpreted their semantic characteristics, applied them to identify landscape scales and urban landscape categories, and explored their potential to reveal underlying patterns, specifically focusing on inter-city and landscape-socioeconomic correlations. The subsequent sections provide a detailed description of this case study.

3.1. Study area and data

This case study involves 36 cities in China, which were selected based on their socioeconomic representativeness, development levels, and geographic locations (see Fig. 2).

Street-level images were obtained through the Tencent Street View Image application programming interface (API).[1] The collection process involved soliciting and capturing street-level images at regular intervals of 50 m along the road networks in 2016. At each street view sampling point, essential information was recorded, including the point ID, its coordinates, and four images captured from distinct angles each with a resolution of 600 × 480 pixels. Overall, a dataset of 8,189,058 images was successfully collected.

To investigate the correlation between urban landscapes and socioeconomic factors, we used urban population, Gross Domestic Product (GDP), and GDP per capita as proxies for the size and development level of a city. Urban population data, with a resolution of 100 m and from the year 2016, were obtained from WorldPop[2] (WorldPop, 2018). Meanwhile, GDP and GDP per capita data were sourced from the 2016 *China Statistical Yearbook* published by the National Bureau of Statistics of China.

3.2. Feature extraction

We process a set of street-level images within a grid, denoted as $\{I_1, I_2, \dots, I_N\}$. Each image I_i undergoes transformation via a scene encoder, producing a feature vector S_i. We utilize the ResNet-18 model (He et al., 2016), a robust architecture that has been widely applied in both computer vision and urban studies (Huang et al., 2023; Wang et al., 2019), as our scene encoder. Specifically, we remove the final fully connected layer and take the output from the global average pooling layer, resulting in a 512-dimensional feature vector for each image. Our choice of ResNet-18 is guided by a balance between model complexity and performance. Compared to deeper variants such as ResNet-50, ResNet-18 offers sufficient representational capacity while maintaining computational efficiency. This trade-off is particularly important in our case study, where a large number of street-level images need to be processed efficiently. While we acknowledge that more complex models may have the potential to capture richer and more nuanced visual representations, our aim in this study is to demonstrate the effectiveness of deep features for urban scene understanding using a lightweight yet representative backbone.

Typically, models pre-trained on natural image datasets (e.g., ImageNet (Krizhevsky et al., 2012)) are widely used in image feature extraction. However, objects in natural images (such as cats and dogs) usually occupy the central position of the image, whereas in street-level images, elements are distributed throughout the entire frame. This distribution necessitates a comprehensive consideration of multiple visual elements' spatial layouts and interrelationships. Consequently, models pre-trained on natural image datasets may not fully adapt to the characteristics of street-level imagery. In this study, the model is pre-trained on the Places 365 dataset—a comprehensive collection of over 10 million images designed for scene classification—making it highly suitable for encoding diverse urban scenes (Zhou et al., 2018).

To obtain the landscape representation from scene features, we calculate both the mean $\overline{S}$ and standard deviation σ of the scene features across the images within a grid. These statistical metrics capture the average condition and intra-landscape variability of the scene features within a grid, respectively. The landscape features L are then formed by concatenating these two metrics. The resulting landscape features are used to identify landscape clusters that share semantic similarities.

3.3. Feature understanding

To investigate the semantic content embedded in deep features and their relevance to downstream tasks, we conducted three interpretability studies using a dataset of over one million street-level images

[1] lbs.qq.com.
[2] www.worldpop.org.

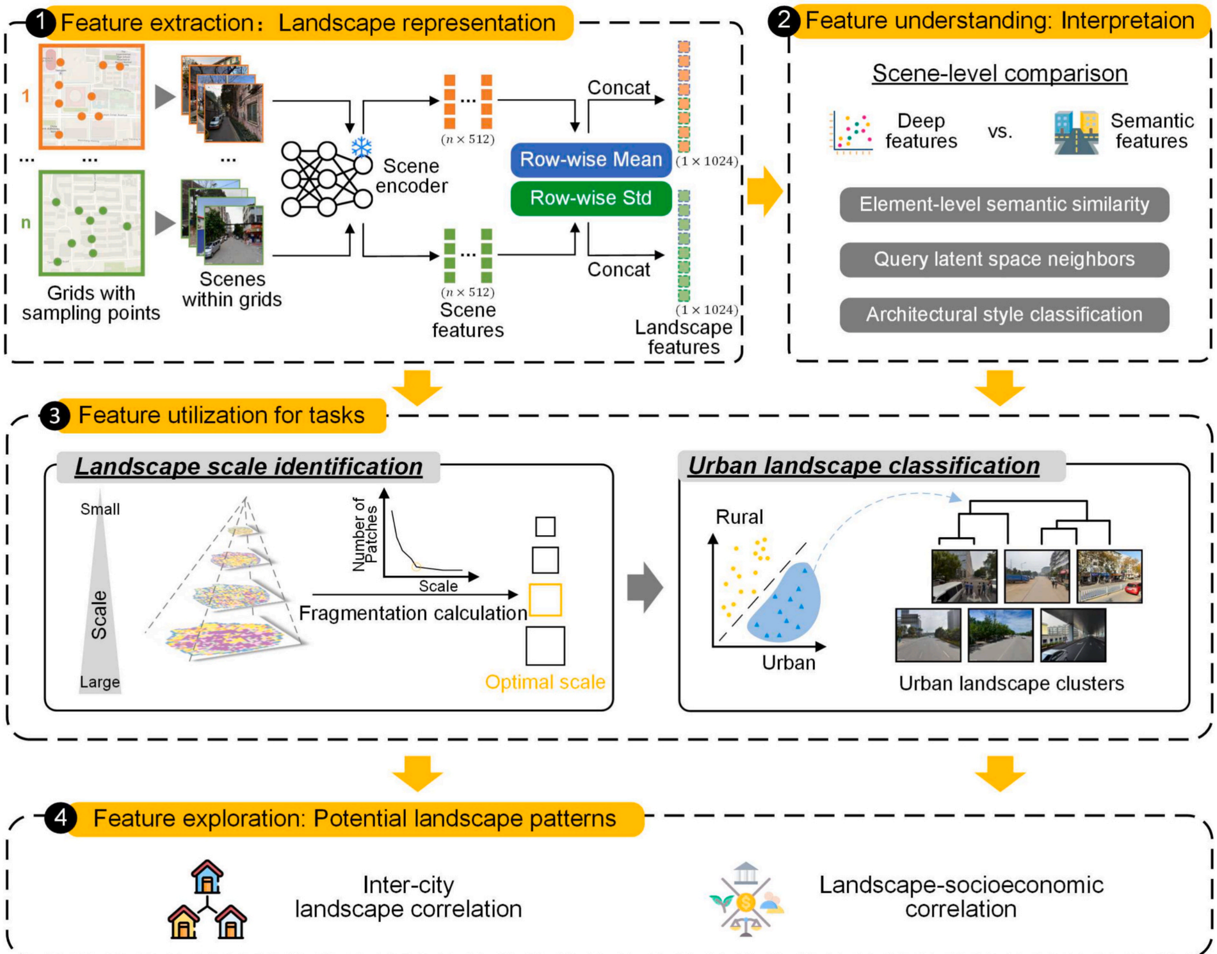

Fig. 1. Overview of the deep feature framework for urban analysis, using urban landscape cell identification as a case study.

from Beijing. These studies compared the information encoded by deep features with that encoded by element-level semantic features. The element-level semantic features were extracted using the Dense Prediction Transformer (DPT) model (Ranftl et al., 2021), pre-trained on the ADE20K dataset (Zhou et al., 2017). This process generated 150-dimensional feature vectors for each image, where each dimension corresponds to the pixel proportion of a specific visual semantic category present in the scene.

To examine whether deep features encode element-level semantic information, we applied the SemAxis method (An et al., 2018). This technique defines semantic axes within the latent space to quantify the degree to which specific semantic labels are embedded in the feature vectors. In our study, the pixel proportions of visual elements obtained from the DPT served as semantic labels. For each label, we defined a semantic axis by calculating the mean deep feature vector of images in the lower quartile (as the start point) and the mean vector of those in the upper quartile (as the end point). All deep features were then projected onto these axes, allowing us to analyze the alignment between element-level semantic features and the latent representations.

To further evaluate the ability of deep features to capture high-level semantic similarity between images, we implemented a nearest neighbor image retrieval task. Specifically, for several randomly selected query images, we computed cosine similarity between the query and all other images using both deep features and element-level semantic features, and retrieved the top-n most similar images for comparison.

Finally, to quantitatively assess the richness of semantic information encoded by deep features, we conducted a classification experiment using architectural style as a case study. Since buildings occupy a significant portion of street-level imagery, the ability to distinguish different architectural styles serves as an indicator of a feature's capacity to represent high-level visual semantics relevant to the urban physical environment. A small-scale dataset was constructed, comprising 150 images across five architectural style categories (30 images per category), with representative samples shown in Fig. 4(c). For each category, 80 % of the images were randomly selected for training and 20 % for testing. Random forest classifiers were trained separately on deep features and semantic features, and the experiment was repeated five times using different random seeds to ensure robustness.

3.4. Feature utilization for tasks

We investigated the utilization of deep features to understand the urban physical environment from two perspectives: landscape scale identification and urban landscape classification.

3.4.1. Landscape scale identification

Street-level imagery provides detailed views of individual street sampling points, capturing diverse urban scenes. As these scenes are combined at varying spatial scales, they reveal different layers of information. At smaller scales, the combined scenes might depict specific local characteristics, such as a basketball court. In contrast, at larger scales, the combination of diverse scenes may reduce clarity, making it challenging to differentiate the information represented by various grids. Thus, identifying an optimal scale where combined scene knowledge effectively represents a cohesive landscape cell is crucial for

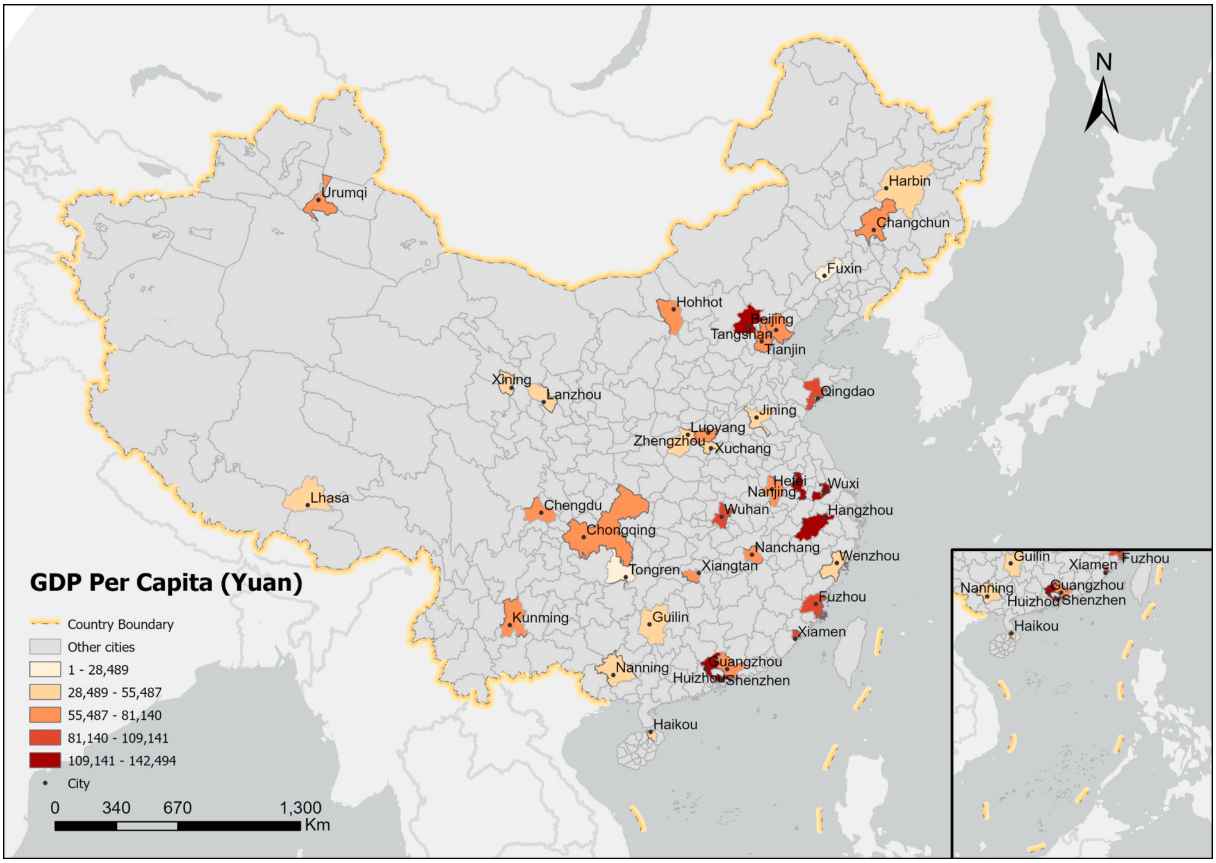

Fig. 2. The 36 cities selected for this study.

accurately deconstructing urban spaces from street-level imagery.

To determine the optimal scale, we began by examining landscape fragmentation. We utilized deep features extracted from the street-level images to perform hierarchical clustering (Bridges, 1966), categorizing landscape cells into a multi-level landscape taxonomy. Cells belonging to the same cluster and that are spatially adjacent were merged to form a landscape patch, which we define as a coherent spatial unit sharing similar latent landscape characteristics. To facilitate this analysis, we employed the Number of Patches (NP), a metric from landscape ecology that quantifies landscape fragmentation, to determine the appropriate size of landscape cells. The NP metric measures the total number of patches within a given area, which can range from one to potentially numerous patches, depending on the complexity of the urban landscape. Generally, a higher NP value indicates greater fragmentation, suggesting a more heterogeneous landscape with diverse urban features.

By evaluating the NP metric across multiple spatial scales, we aimed to identify the scale at which fragmentation reaches a plateau. This plateau indicates that increasing the spatial scale further does not significantly change the level of fragmentation, implying that the scene knowledge at this scale effectively encompasses relatively complete landscape characteristics. In other words, this optimal scale represents a balance where the aggregated deep features provide a coherent and meaningful representation of the urban landscape without losing essential local details.

3.4.2. *Urban landscape classification*

It is crucial to understand that the street view sampling areas do not necessarily correspond to urban areas. Fig. 3 illustrates the relationship

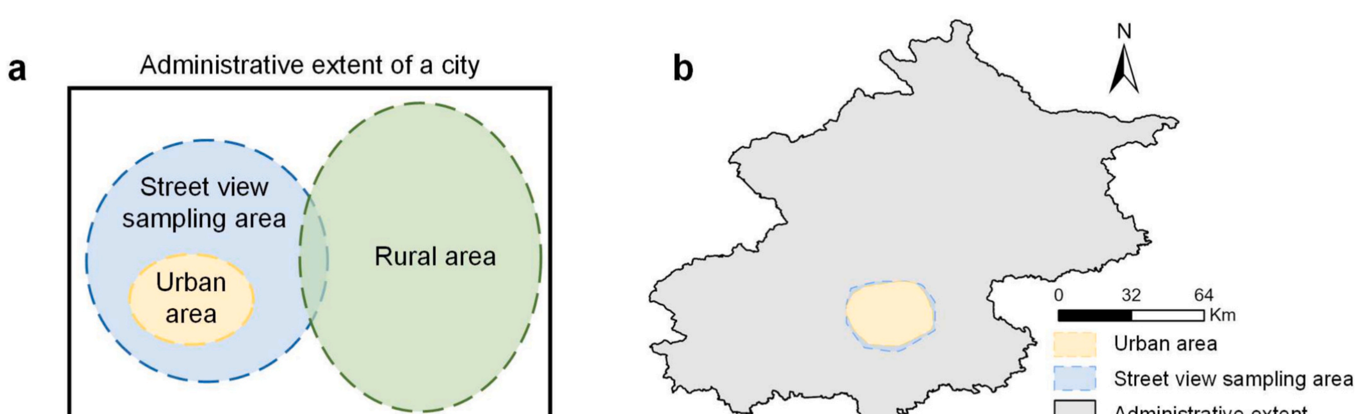

Fig. 3. Relationship between administrative extent, street view sampling area, and urban area. (a) A conceptual city; (b) Beijing.

between administrative extents, street view sampling areas, and urban areas. Currently, there is no universally accepted ground truth that precisely defines urban areas, and the definition itself remains subject to interpretation (Cao et al., 2023).

From a landscape perspective, there is often a clear distinction between urban and rural environments based on physical characteristics and land use patterns. In this study, we aim to classify the landscapes within the street view sampling areas to effectively distinguish urban from rural environments and to identify fundamental urban landscape categories. To achieve this, we first performed hierarchical clustering on the deep features extracted from the street-level images within the sampling areas. Initially, we classified the landscapes into two clusters to segregate urban and rural areas based on their deep feature representations.

To validate and label these clusters appropriately, we integrated population data from WorldPop. By associating each cluster with its corresponding population density, we identified the cluster with a higher average population density as the "urban" cluster and the one with lower density as the "rural" cluster. This integration of socioeconomic data helps ensure that our classification aligns with practical definitions of urban areas and provides a more robust distinction between urban and rural landscapes.

With the urban landscapes identified, we proceeded to further classify the features of these urban landscapes to uncover fundamental urban landscape categories across all cities studied. Again, we employed hierarchical clustering on the deep features of the urban landscape cells. The number of clusters in hierarchical clustering directly affects the granularity of the classification: using a higher number of clusters allows for a finer division of landscape categories, while a smaller number results in more generalized classifications. Our objective is to determine the optimal number of clusters that best represent the fundamental urban landscape categories. To do this, we utilized the cluster distance as a key metric. In hierarchical clustering, cluster distance measures the proximity between clusters and guides the decisions on merging or dividing clusters at each step of the algorithm.

3.5. Feature exploration

To uncover deeper potential patterns within the urban physical environment, we explored the deep features through two analyses: intercity landscape correlation and landscape-socioeconomic correlation. Such analyses were designed to enhance the understanding of how deep features capture complex urban characteristics.

We explored similarities between cities by analyzing the deep features of urban landscape cells. While urban landscapes can be categorized into fundamental types, factors such as policies and cultural influences may cause specific landscape types within certain urban clusters to exhibit similar characteristics. To capture this potential trend, we calculated the average feature distances between different cities' landscape cells in the latent space, providing a quantitative measure of overall similarity based on the deep features extracted from street-level images.

Recognizing that landscapes within the same city can also be diverse, we applied dimensionality reduction to project the high-dimensional deep features onto a lower-dimensional space for visualization. Specifically, we used PaCMAP (Wang et al., 2021), a technique that preserves both global and local structures in high-dimensional data. By projecting the deep features onto two dimensions, we were able to visualize relationships among landscape cells and identify patterns and clusters not apparent in the high-dimensional latent space.

Moreover, to examine the relationship between urban landscapes and socioeconomics, we analyzed correlations between the proportions of landscape types in each city and various socioeconomic indicators. This analysis aimed to validate the effectiveness of deep features in capturing meaningful urban patterns by highlighting the interplay between urban landscapes and socioeconomic development.

4. Results

4.1. Interpretation of deep features

To understand what information is encoded within the deep features, we conducted three interpretability analyses. Firstly, we investigated whether deep features encode information related to visual elements by applying the SemAxis method to explore the relationship between feature representations and semantic content. Based on the results of semantic segmentation from DPT, the top 20 visual elements with the highest average pixel proportions were identified. These elements were used as semantic labels in the SemAxis analysis, where their pixel proportions served as the basis for defining semantic axes. As shown in Fig. 4(a), the analysis indicates that the pixel proportions of these visual elements exhibit significant positive correlations ($p < 0.001$) with their projection values on the corresponding semantic axes defined by SemAxis. This demonstrates that the deep features can effectively capture these prominent visual elements. Furthermore, we observed that the higher the pixel proportion of a visual element in the images, the stronger the correlation between its proportion and its projection value on the semantic axis. This may be because visual elements with higher proportions have more consistent and salient representations in the latent space, leading to stronger correlations in the SemAxis projections.

To further investigate the high-level semantic information encoded in deep features, we conducted a nearest neighbor image retrieval experiment, as illustrated in Fig. 4(b). For three randomly selected query images, we retrieved their nearest neighbors in both the semantic space constructed from element-level semantic features and the latent space derived from deep features. In the semantic feature space, the retrieved images exhibited highly similar pixel-level distributions of visual elements—for instance, many predominantly contained buildings. However, despite these surface-level similarities, the actual semantic content varied substantially across images. In contrast, the deep feature-based latent space yielded nearest neighbors that not only resembled the query images in visual appearance but also aligned more closely in terms of semantic meaning, such as architectural style or urban scene context. This suggests that deep features capture richer and more nuanced representations of urban environments. To quantitatively validate this finding, we conducted an architectural style classification task using five distinct categories (see Fig. 4(c)). The deep feature-based model achieved perfect classification accuracy (100 %), while the model based on semantic features achieved a considerably lower average accuracy of 79.33 %. This result indicates that semantic features alone are insufficient to discriminate between architectural styles and that their classification performance likely depends on auxiliary categorical cues rather than on the visual semantics themselves.

4.2. Identifying the optimal landscape scale using deep features

To determine the optimal landscape scale for our analysis, we calculated the landscape features of 36 cities using grid sizes of 300 m, 500 m, 700 m, 1000 m, 1500 m, and 2000 m. At each scale, we aggregated the deep features within the grids and conducted landscape clustering analysis based on these aggregated landscape features. Taking Beijing as a visualization example, when classifying the landscapes into ten clusters, the spatial distribution of different landscape types is shown in Fig. 5(a). The results indicate that smaller grid sizes lead to more fragmented landscapes, while larger grids (e.g., 1500 m and 2000 m) include more diverse scenes within a single grid cell, reducing distinctions between cells and forming larger, more homogeneous landscape patches.

As illustrated in Fig. 5(b), we calculated the number of landscape patches at each scale across all 36 cities, revealing a decreasing trend in fragmentation as grid size increases. Notably, there is a significant slowdown in this trend at the 1000 m grid size, indicating an inflection point. This suggests that at the 1000 m grid size, the landscape cells

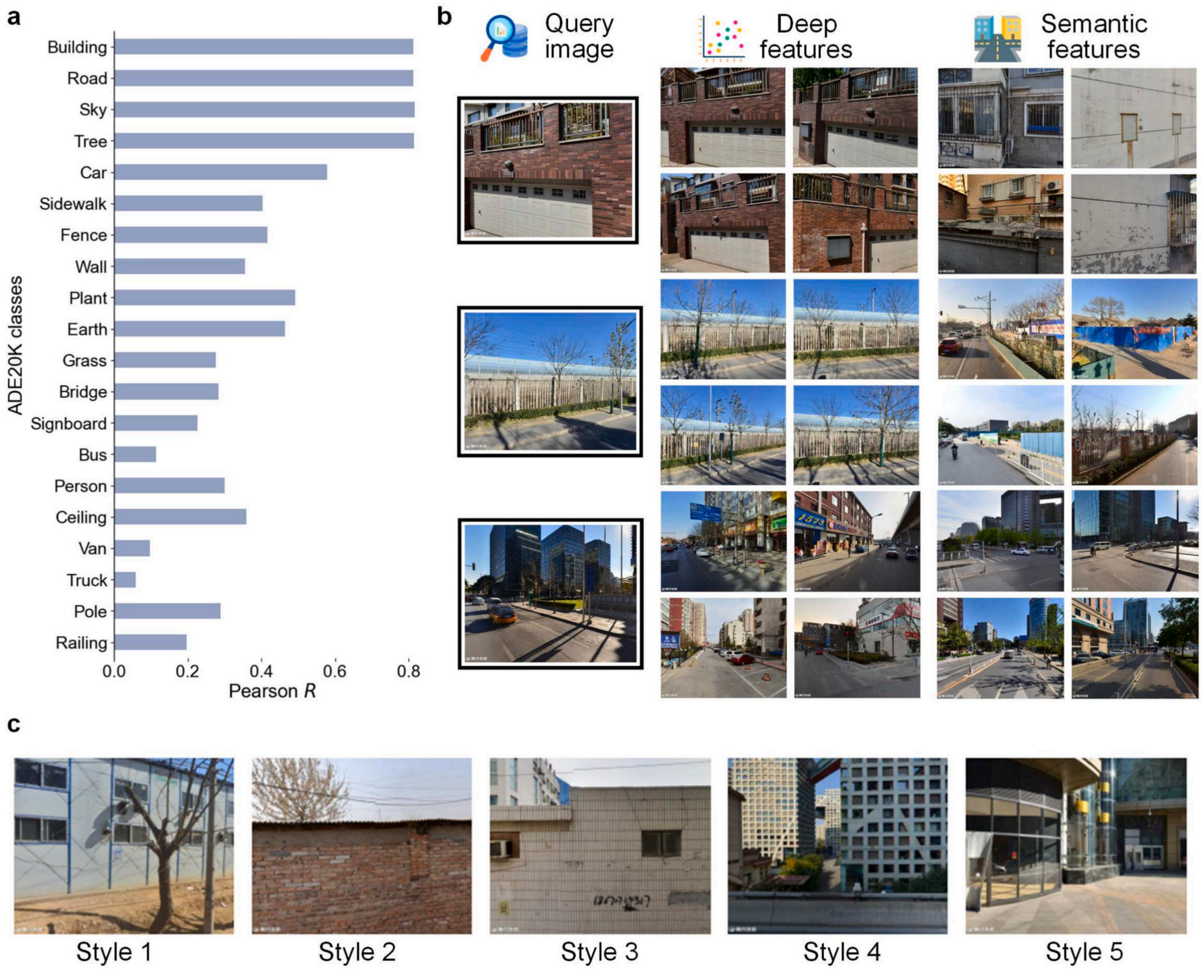

Fig. 4. Interpretation of deep features. (a) Correlation analysis using the SemAxis method, illustrating the relationship between the pixel proportions of visual elements and their corresponding SemAxis projection values. The plot highlights the top 20 visual element categories with the highest pixel proportions. All correlations are statistically significant at $p < 0.001$; (b) Nearest neighbor images retrieved based on a query image in the visual element-based semantic space and deep feature-based latent space; (c) Representative images of the five architectural style categories used in the classification experiment. Each category contains 30 images, selected to illustrate the diversity of styles and support the evaluation of feature-based classification models.

reach an optimal balance where fragmentation is minimized without overly generalizing the urban scenes. Therefore, we selected the 1000 m grid size as the standard landscape cell size for our subsequent analyses. This optimal scale supports the capability of deep features to capture the essential characteristics of the urban physical environment without losing critical details due to over-aggregation or fragmentation.

4.3. Classification of urban landscape categories using deep features

To focus our analysis specifically on urban landscapes and distinguish them from non-urban areas, we first employed clustering to categorize the initial landscape cells into two distinct groups, thereby identifying the urban subset for subsequent detailed study. Based on population density from WorldPop, we designated the cluster with higher population density as the urban area. As illustrated in Fig. 6, four cities of varying sizes and geographical locations—Xining, Haikou, Wuhan, and Tianjin—all exhibit a clear urban-rural distinction. Notably, this method is effective for both monocentric and polycentric cities. For example, Tianjin, a typical bicentric city, has both its traditional city center and the newly planned Binhai New Area accurately identified. Moreover, the urban landscapes of these cities are distinctly different from rural landscapes, which typically feature undeveloped land and traditional village houses. In contrast, urban areas are characterized by green belts, high-rise buildings, and elevated bridges.

To determine the fundamental landscape categories, we performed hierarchical clustering on the landscape features of urban areas. As shown in Fig. 7(a), we observed a general downward trend in cluster distances. There is a rapid decline from two to five clusters, followed by a more gradual decrease after six clusters. Based on these findings, we inferred that urban landscapes primarily consist of six fundamental categories. Fig. 7(b) displays the hierarchical clustering dendrogram for these six categories, each annotated with its visual characteristics. The six identified landscape categories are as follows:

Daily Life Commercial District: Characterized by a variety of architectural styles and bustling activity, this landscape features narrow streets, often with roadside parking. Small-scale shops line these streets, with many ground-floor stores offering daily necessities. These elements reflect the daily rhythms and consumption patterns of urban residents.

Transportation Hub: Distinguished by a concentration of diverse transportation facilities, this landscape includes overpasses, traffic circles, and broad multi-lane roads. The area is equipped with advanced traffic management systems, such as traffic lights and road signs, and is defined by spacious roads and open spaces that often include roundabouts or bridge structures. Typically, commercial buildings or office towers are located nearby.

Old Town: This landscape comprises industrial zones and low-rise buildings. Medium to large trucks are frequently seen, and the surrounding buildings are primarily single-story, flat-roofed structures with

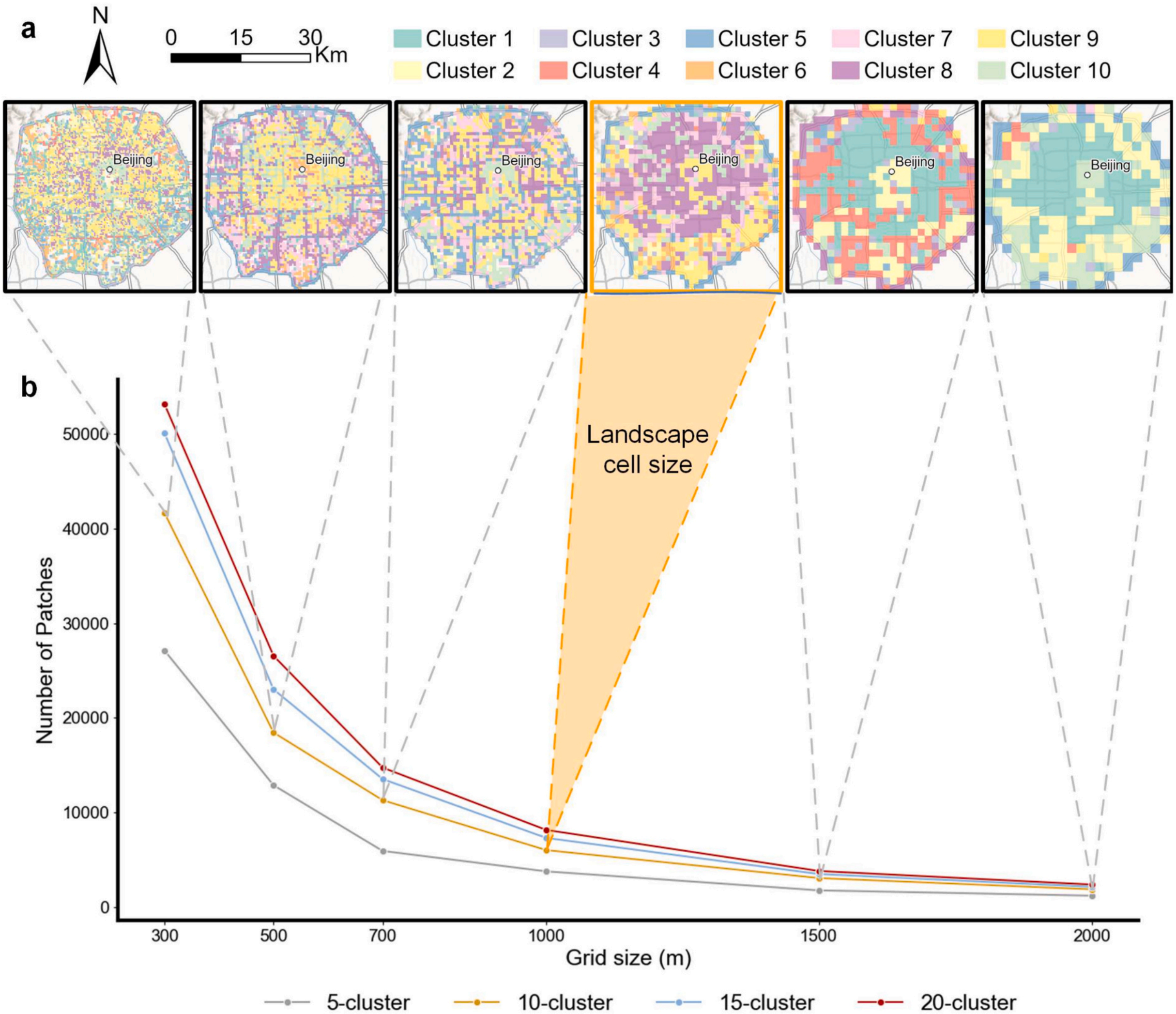

Fig. 5. Landscape fragmentation with varying grid sizes. (a) Spatial distribution of Beijing's landscape patches based on a 10-cluster classification at different grid sizes; (b) The number of landscape patches across all 36 cities at different grid sizes, illustrating the trend in fragmentation.

high density. Notably, this area lacks greenery, giving it a more utilitarian appearance.

Green Space: Dominated by lush greenery along the streets, this landscape offers wide roads with fewer surrounding buildings. High-rise buildings may be visible in the distance, creating a backdrop suitable for leisure and walking with relatively low traffic volumes.

Modern Business Center: This landscape is dominated by modern high-rise buildings with bright glass curtain walls and contemporary design elements. The street layout is orderly, reflecting the commercial and cultural dynamism of the urban core.

Thoroughfare: Characterized by wide roads flanked by modern high-rise buildings, this landscape features buildings that are spaced relatively far apart due to the breadth of the roads. The architectural style is modern, supporting the landscape's function as a major urban thoroughfare.

Taking Chengdu and Guangzhou as examples, as shown in Fig. 7(c) and (d), we observe that the spatial distribution of landscape patterns can vary significantly across cities. In Chengdu, a large number of Daily Life Commercial Districts are concentrated in the central area, while Modern Business Centers are located in the peripheral regions. In contrast, in Guangzhou, although the central area contains some Daily Life Commercial Districts, Modern Business Centers are also situated near the central area. Green Spaces are mostly distributed in peripheral areas; however, landscapes near parks are often categorized as Green Space regardless of their proximity to the city center. These variations highlight how different urban development strategies and historical factors influence the spatial arrangement of landscape categories within cities. The ability of deep features to capture these nuances demonstrates their effectiveness in classifying and analyzing urban landscapes.

4.4. Inter-city landscape correlations revealed by deep features

To investigate whether inter-city relationships could be inferred from landscape similarity, we calculated the average distances between the landscape cells of each city and those of other cities within the latent space corresponding to specific urban landscape types. This approach leverages deep features to reveal latent patterns and structural similarities across cities. Using the Modern Business Center category as an example (Fig. 8(a)), we observe a clear clustering trend among cities. Most cities display similar landscape characteristics within this category, particularly those located in the lower region of the figure—namely, Wenzhou, Fuzhou, Kunming, Chengdu, Hangzhou, Hefei, Wuxi, and Huizhou. Guilin, in particular, demonstrates high internal consistency and strong resemblance to all cities. By contrast, Guangzhou exhibits a distinct profile compared to most other cities but forms a small, coherent cluster with Chongqing and Shenzhen. Additionally, Harbin, Urumqi, and Changchun constitute another separate cluster. These groupings suggest that cities within each cluster share common

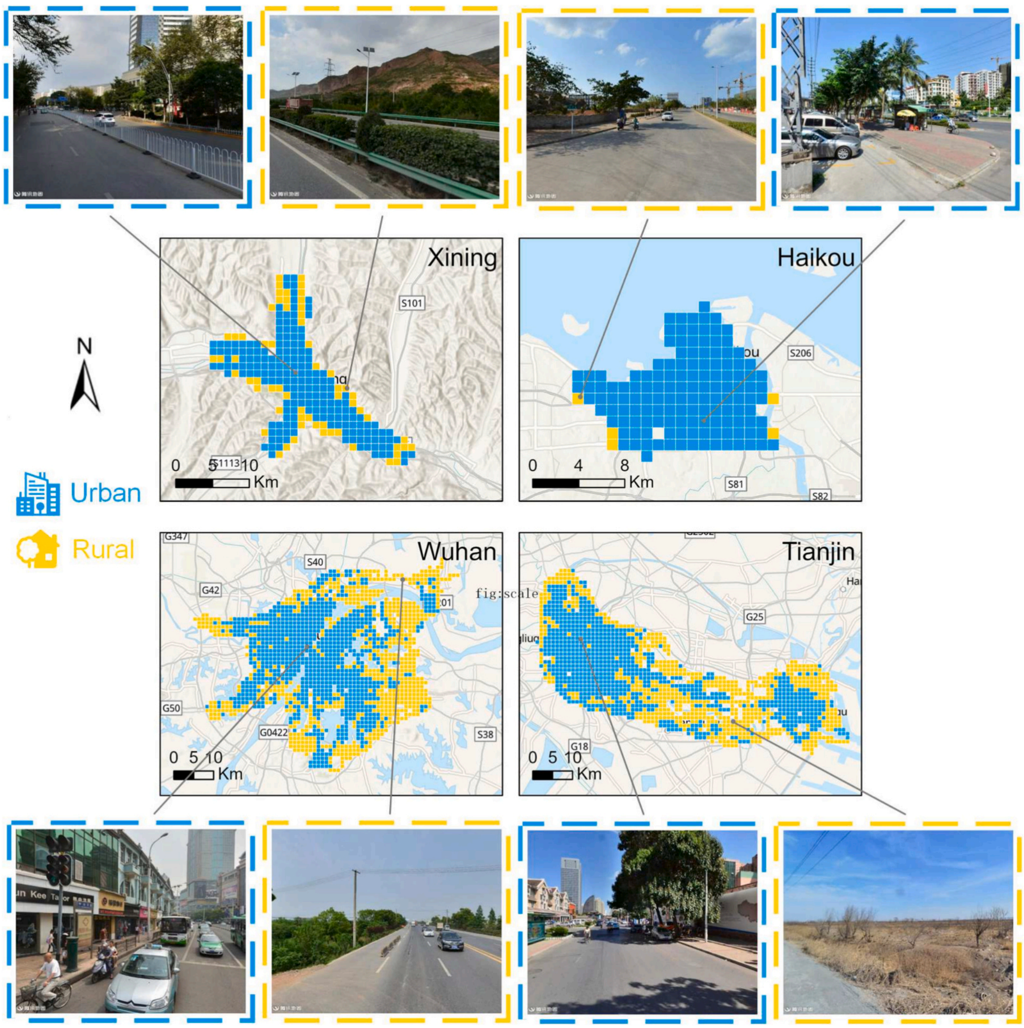

Fig. 6. Extraction of urban and rural areas based on landscape features. Four cities are presented as examples: Xining, Haikou, Wuhan, and Tianjin.

Modern Business Center landscape features, potentially reflecting similar urban development trajectories or underlying geographical and socioeconomic conditions.

By projecting the high-dimensional deep features onto a two-dimensional space using PaCMAP, we can visualize the relationships among landscape cells more effectively. As illustrated in Fig. 8(b), the PaCMAP-based dimensionality reduction reveals that cities such as Chongqing, Guangzhou, Hangzhou, Beijing, and Haikou form five distinct clusters at the peripheries of the feature space. These clusters exhibit significant spatial separation, indicating unique urban characteristics in their Modern Business Center landscapes. These findings suggest that certain urban characteristics are prominently reflected in specific landscape cells rather than uniformly across all cells. The distinct clusters identified through PaCMAP highlight the diversity of urban landscapes among different cities. For example, the clustering of Chongqing, Guangzhou, and Shenzhen may reflect similarities in their rapid economic development and modern urban design, while the cluster formed by Harbin, Urumqi, and Changchun may be influenced by their northern geographical locations and climatic conditions. This analysis underscores the ability of deep features to uncover latent correlations in urban landscapes, providing deeper insights into the complex spatial patterns across cities.

4.5. Landscape-socioeconomic correlations revealed by deep features

To explore whether urban landscape types are influenced by, or in turn influence, socioeconomic factors, we conducted a correlation analysis between the proportions of specific landscape types and various socioeconomic indicators. As shown in Fig. 9(a), the proportion of Daily Life Commercial Districts exhibits a significant negative correlation with all three socioeconomic indicators: total GDP, GDP per capita, and urban population. This suggests that cities with higher populations and greater economic development tend to have fewer street-level commercial landscapes. Conversely, the proportion of Transportation Hubs shows a significant positive correlation with urban population and GDP, indicating that larger and more economically developed cities have more extensive transportation infrastructure landscapes.

Fig. 9(b) illustrates the negative correlation between the proportion of Daily Life Commercial Districts and GDP per capita, with a Pearson correlation coefficient of -0.5210 ($p < 0.01$). This indicates that cities with higher GDP per capita tend to have fewer street-level commercial landscapes. One possible explanation is that in wealthier cities, large shopping malls and commercial centers have supplanted traditional street-level commerce, centralizing consumer services and reducing the prevalence of small-scale commercial districts. For example, in Beijing, such street-level commercial landscapes are relatively scarce, highlighting the city's reliance on larger commercial complexes for daily consumer needs.

Fig. 9(c) shows the positive relationship between the proportion of Transportation Hubs and urban population, with a Pearson correlation coefficient of 0.7026 ($p < 0.001$). This suggests that cities with larger populations require more comprehensive transportation systems,

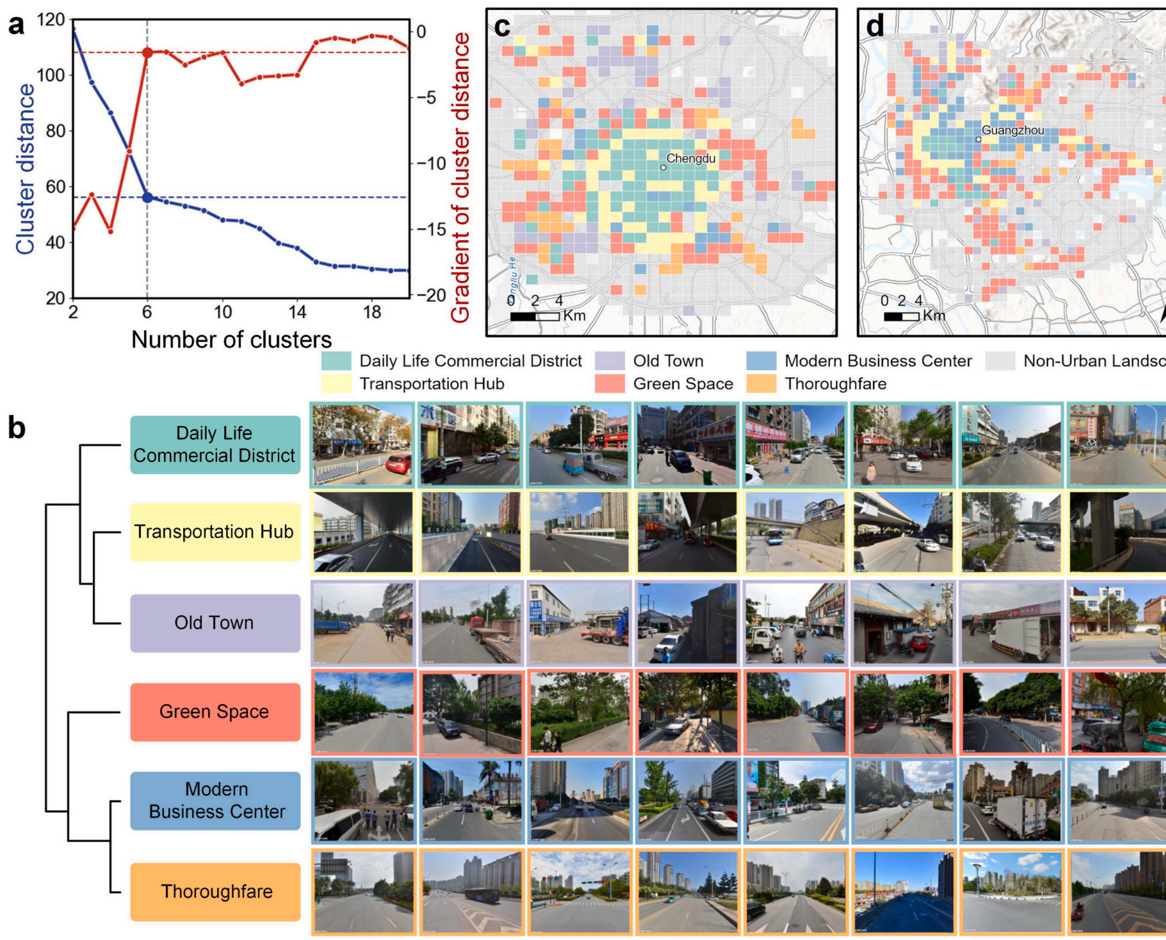

Fig. 7. Taxonomy of urban landscapes. (a) Changes in cluster distances indicating the number of fundamental landscapes; (b) Hierarchical clustering dendrogram with annotations of visual characteristics; (c) Landscape distribution in Chengdu; (d) Landscape distribution in Guangzhou.

including features such as pedestrian overpasses and traffic signal networks, to facilitate efficient mobility. Interestingly, Shenzhen, despite its large population, has relatively fewer landscapes categorized as Transportation Hubs. This anomaly may be attributed to the city's high industrial density and urban planning strategies that concentrate populations in specific areas such as Huaqiangbei, thereby reducing the need for widespread transportation infrastructure compared to other large cities. Conversely, smaller cities such as Xiangtan, Lhasa, Fuxin, and Guilin, with lower population densities, lack extensive transportation hub landscapes due to less demand for such infrastructure.

These findings demonstrate that deep features can effectively reveal latent correlations between urban landscapes and socioeconomic factors, providing valuable insights into how the physical environment relates to economic and demographic characteristics.

5. Discussion

5.1. Advancing urban landscape characterization

The study demonstrates the effectiveness of deep features for identifying urban landscape cells. By providing a standardized method to describe urban landscape features, this approach facilitates cross-city comparisons and offers new perspectives for urban science research. The identification of an optimal landscape scale contributes to urban studies by providing a data-driven basis for defining spatial analysis units. Instead of relying on administratively defined boundaries, our fragmentation-based approach reveals the scale at which coherent landscape patterns emerge. This insight can support more accurate spatial modeling. Compared to segmentation-based clustering approaches such as Liang et al. (2023), which rely on predefined semantic labels, our method captures more composite and nuanced landscape structures derived from visual co-occurrence and spatial texture. For example, their cluster Urban Jungle merges visually distinct areas such as high-rise commercial centers and informal self-built neighborhoods, whereas our method is able to differentiate more clearly. By leveraging high-dimensional feature embeddings, our framework enables the identification of latent structural differences between urban landscapes. Moreover, the semantic richness of deep features not only supports a more detailed landscape classification but also allows for further analysis within each landscape type. This includes uncovering inter-city similarities and differences, as well as exploring correlations between landscape patterns and socioeconomic conditions—thereby providing insights into the spatial logic of urban development.

While street-level imagery primarily captures urban street spaces and offers limited visibility into the interiors of street blocks, the visual characteristics of streets often reflect the nature of adjacent urban areas. For instance, Huang et al. (2023) demonstrated that the presence of informal residential areas within blocks could be identified using street-level imagery alone, achieving an accuracy of 82.8 %. This finding highlights that, despite inherent limitations, street-level imagery remains a valuable source for extracting meaningful insights about the broader urban environment. These results support the utility of our approach in leveraging visual cues from streetscapes to infer broader patterns of urban form and function.

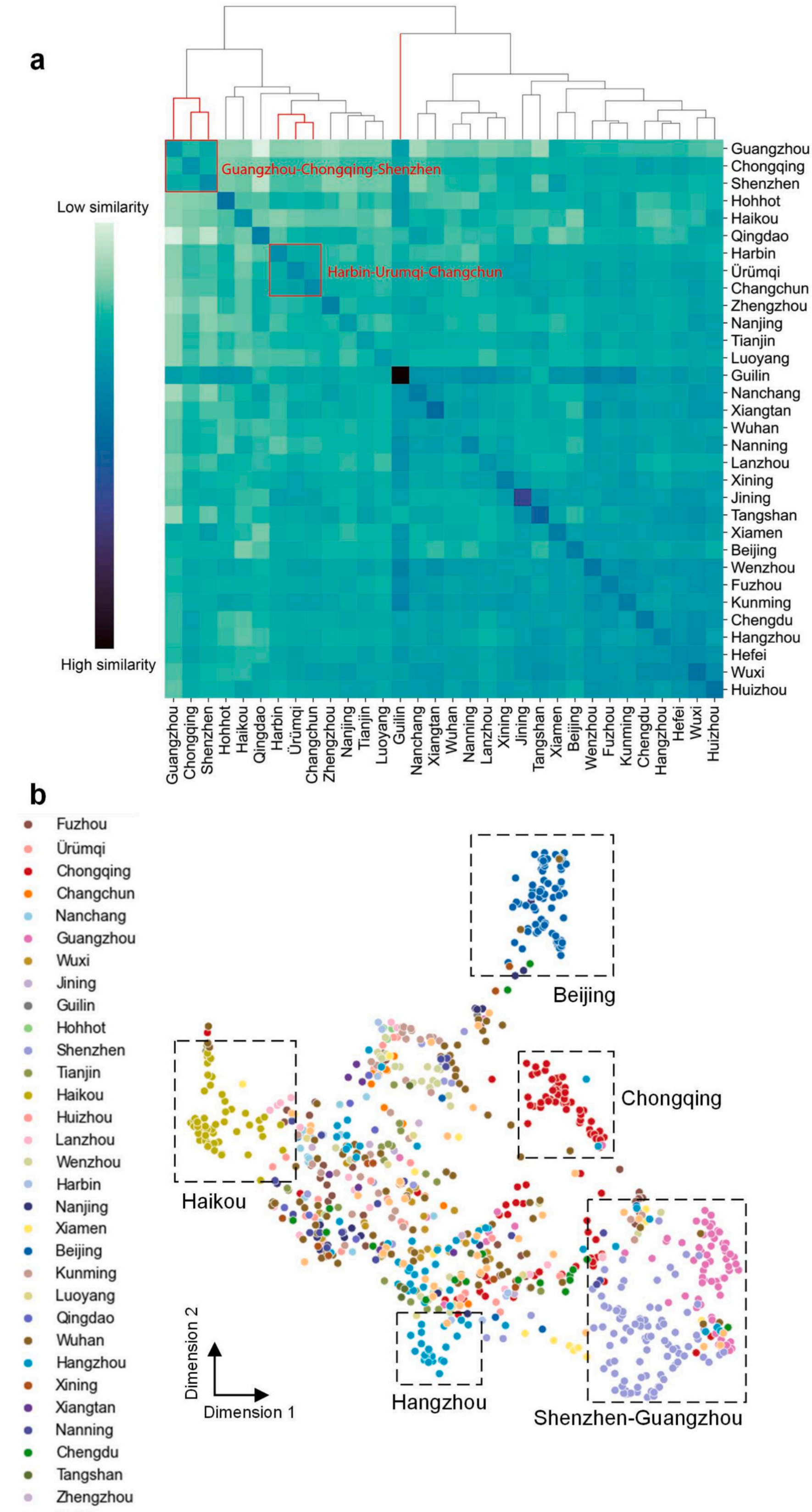

Fig. 8. Clustering patterns within the latent space of Modern Business Center. (a) Overall clustering pattern of cities, illustrated using a heatmap based on the standardized distance matrix, where darker colors indicate higher similarity and lighter colors indicate lower similarity; (b) Detailed clustering pattern of cities visualized using PacMAP.

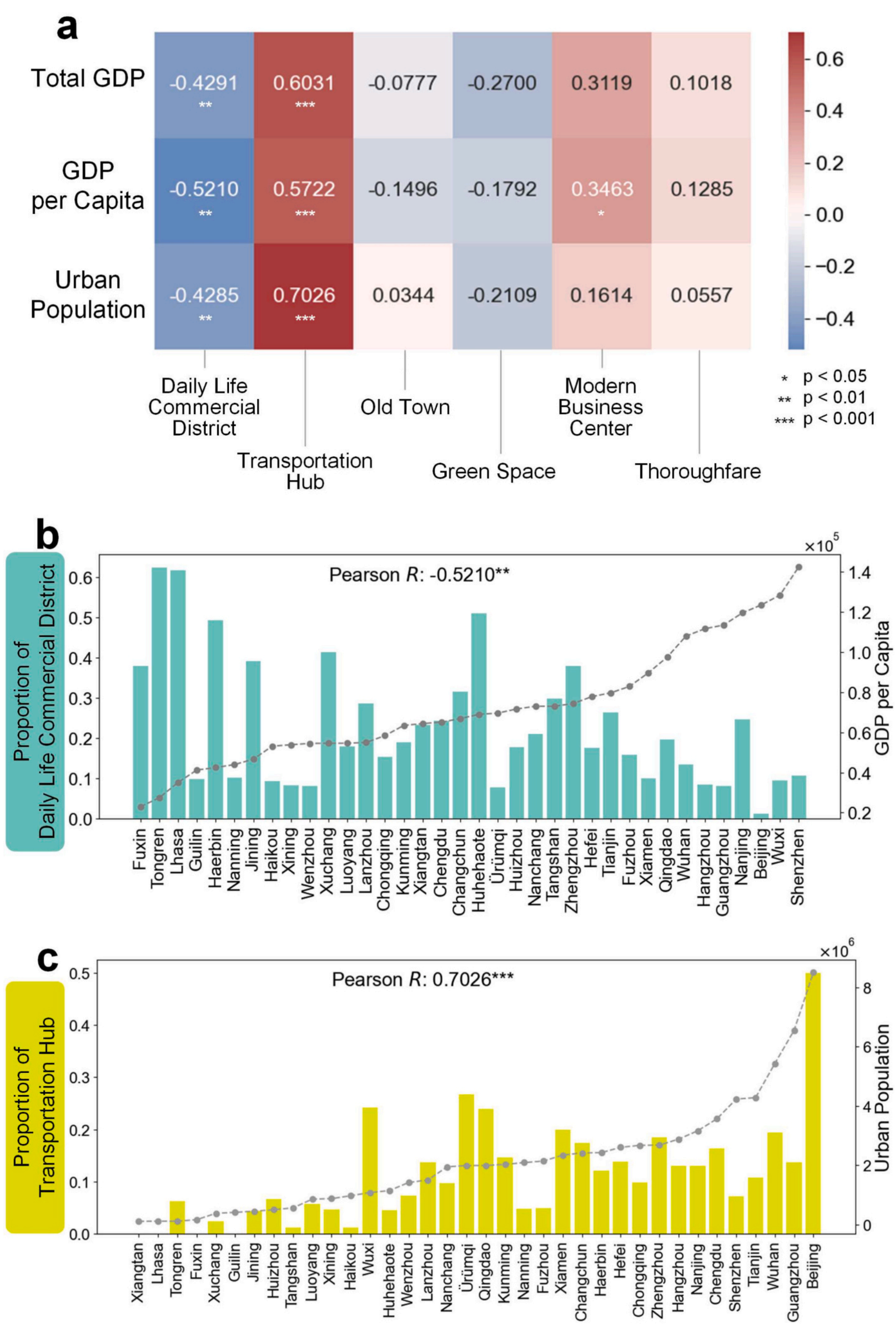

Fig. 9. Correlation between urban landscapes and socioeconomic factors. (a) Pearson correlation coefficients between six types of urban landscapes and three socioeconomic indicators: total GDP, GDP per capita, and urban population; (b) Relationship between the proportion of Daily Life Commercial Districts and GDP per capita across cities; (c) Relationship between the proportion of Transportation Hubs and urban population across cities.

5.2. Implications for urban science and practice

The significance of deep features in urban science is manifested mainly in the following aspects. First, it provides a more powerful tool for understanding the relationships between the urban physical environment and other urban layers such as socioeconomics, environmental quality, and transportation patterns. By uncovering hidden patterns and deep correlations, computational representation offers scientific support for urban planning and policy-making. For instance, by comparing deep feature-derived landscape patterns with zoning plans, our approach can help detect spatial development inconsistencies, such as misaligned land uses or fragmented urban growth, enabling planners to prioritize interventions. Second, it enhances the comparability of cross-city studies and the transferability of results. The standardized approach to urban representation allows for efficient comparison of physical environment characteristics across different cities, thereby promoting collaborative research and the sharing of best practices. Moreover, deep features create new possibilities for the integration and analysis of multimodal data. By combining image deep features with other data sources, such as points of interest (POIs) data, researchers can construct representations of the urban environment from multiple perspectives, providing richer support for the development and practical innovation of urban science.

5.3. Limitations and future works

While different backbone architectures may yield slight variations in feature extraction performance, we consider the quality and composition of the pre-training dataset to have a more substantial impact on the downstream effectiveness of scene representations. Recent studies (Li et al., 2025) have shown that pre-training on datasets specifically curated for urban contexts leads to more semantically meaningful and task-relevant features in urban applications. As large-scale foundation models continue to evolve, incorporating not only vast visual datasets but also structured world knowledge and context-aware semantic understanding (Hou et al., 2025), future work may explore how such models can further enhance the expressiveness, generalizability, and interpretability of urban scene representations.

In this study, we used a regular grid structure to aggregate street-level image features. This approach offers simplicity and consistency across different urban areas, making it suitable for large-scale comparative analysis. However, we acknowledge that alternative spatial units, such as H3 hexagonal grids or street-block-based divisions, may provide more context-aware representations by aligning better with the underlying urban morphology or road network structure.

This study focuses exclusively on Chinese cities, which may limit the generalizability of the findings to other geographic and cultural contexts. While the proposed methodology is transferable in principle, applying it to cities in different countries may require adaptation to account for variations in urban morphology, imagery characteristics, or landscape semantics. In addition, although deep features offer the advantage of capturing rich latent visual information, they inherently lack transparency due to their black-box nature. Future work may benefit from incorporating more explainable feature representations or hybrid approaches that combine deep features with interpretable semantic cues.

6. Conclusion

This study proposed a framework that leverages deep features to characterize urban physical environments and support urban analysis. By utilizing large-scale street-level imagery, the framework encodes rich visual information into high-dimensional representations, capturing structural patterns and spatial complexity that are often overlooked by traditional, manually defined semantic features. Through a case study on urban landscape cell identification across 36 Chinese cities, we've shown that deep features enable fine-grained landscape classification, robust spatial pattern analysis, and meaningful inter-city comparisons. This work ultimately opens new avenues for data-driven urban research, providing a powerful foundation for understanding and addressing the multifaceted challenges of urban environments.

CRediT authorship contribution statement

Yingjing Huang: Writing – review & editing, Visualization, Investigation, Conceptualization, Writing – original draft, Validation, Software, Methodology, Formal analysis. **Fan Zhang:** Funding acquisition, Supervision, Methodology, Validation, Resources, Data curation, Writing – review & editing. **Lun Wu:** Validation, Writing – review & editing. **Yu Liu:** Project administration, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no conflict of interest.

Acknowledgement

This work was supported by the National Natural Science Foundation of China under Grant 42371468. This work was also supported by High-performance Computing Platform of Peking University. The authors would like to thank Yong Li, Furong Jia, and Ce Hou for their valuable suggestions.

Data availability

The deep features and result labels are accessible at https://doi.org/10.5281/zenodo.13165329.

References

An, J., Kwak, H., & Ahn, Y.-Y. (2018). Semaxis: A lightweight framework to characterize domain-specific word semantics beyond sentiment. In *Proceedings of the 56th annual meeting of the Association for Computational Linguistics (volume 1: Long papers)* (pp. 2450–2461). Melbourne: Australia. Association for Computational Linguistics.

Arnheim, R. (1954). *Art and visual perception: A psychology of the creative eye*. Berkeley: University of California Press.

Biljecki, F., & Ito, K. (2021). Street view imagery in urban analytics and gis: A review. *Landscape and Urban Planning, 215*, Article 104217.

Branas, C. C., South, E., Kondo, M. C., Hohl, B. C., Bourgois, P., Wiebe, D. J., & MacDonald, J. M. (2018). Citywide cluster randomized trial to restore blighted vacant land and its effects on violence, crime, and fear. *Proceedings of the National Academy of Sciences, 115*(12), 2946–2951.

Bridges, C. C. (1966). Hierarchical cluster analysis. *Psychological Reports, 18*(3), 851–854.

Cao, W., Dong, L., Cheng, Y., Wu, L., Guo, Q., & Liu, Y. (2023). Constructing multi-level urban clusters based on population distributions and interactions. *Computers, Environment and Urban Systems, 99*, Article 101897.

Chen, X., Fan, H., Girshick, R., & He, K. (2020). *Improved baselines with momentum contrastive learning*.

Cordts, M., Omran, M., Ramos, S., Rehfeld, T., Enzweiler, M., Benenson, R., Franke, U., Roth, S., and Schiele, B. (2016). The cityscapes dataset for semantic urban scene understanding. In 2016 IEEE conference on computer vision and pattern recognition (CVPR), Las Vegas, NV, USA.

Dalal, N., & Triggs, B. (2005). Histograms of oriented gradients for human detection. In *, 1. Computer vision and pattern recognition, 2005. CVPR 2005. IEEE computer society conference on* (pp. 886–893). IEEE.

Doersch, C., Gupta, A., & Efros, A. A. (2015). Unsupervised visual representation learning by context prediction. In *2015 IEEE international conference on computer vision (ICCV)* (pp. 1422–1430).

Dou, P., & Chen, Y. (2017). Dynamic monitoring of land-use/land-cover change and urban expansion in Shenzhen using landsat imagery from 1988 to 2015. *International Journal of Remote Sensing, 38*(19), 5388–5407.

Fan, Z., Zhang, F., Loo, B. P. Y., & Ratti, C. (2023). Urban visual intelligence: Uncovering hidden city profiles with street view images. *Proceedings of the National Academy of Sciences, 120*(27), Article e2220417120.

Gebru, T., Krause, J., Wang, Y., Chen, D., Deng, J., Aiden, E. L., & Fei-Fei, L. (2017). Using deep learning and google street view to estimate the demographic makeup of neighborhoods across the United States. *Proceedings of the National Academy of Sciences, 114*(50), 13108–13113.

Guo, Z., & Loo, B. P. (2013). Pedestrian environment and route choice: Evidence from New York city and Hong Kong. *Journal of Transport Geography, 28*, 124–136.

He, K., Fan, H., Wu, Y., Xie, S., & Girshick, R. (2020). Momentum contrast for unsupervised visual representation learning. In *2020 IEEE/CVF conference on computer vision and pattern recognition (CVPR)* (pp. 9726–9735).

He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *2016 IEEE conference on computer vision and pattern recognition (CVPR)* (pp. 770–778).

Hoedl, S., Titze, S., & Oja, P. (2010). The bikeability and walkability evaluation table. *American Journal of Preventive Medicine, 39*(5), 457–459.

Hou, C., Zhang, F., Kang, Y., Gao, S., Li, Y., Duarte, F., & Li, S. (2024). Transferred bias uncovers the balance between the development of physical and socioeconomic environments of cities. *Annals of the American Association of Geographers, 115*(1), 148–166.

Hou, C., Zhang, F., Li, Y., Li, H., Mai, G., Kang, Y., Yao, L., Yu, W., Yao, Y., Gao, S., Chen, M., & Liu, Y. (2025). Urban sensing in the era of large language models. *The Innovation, 6*(1), Article 100749.

Howard, E. (1902). *Garden city of tomorrow*. London: Passim.

Huang, Y., Sanatani, R. P., Liu, C., Kang, Y., Zhang, F., Liu, Y., … Ratti, C. (2025). No "true" greenery: Deciphering the bias of satellite and street view imagery in urban greenery measurement. *Building and Environment, 269*, Article 112395.

Huang, Y., Zhang, F., Gao, Y., Tu, W., Duarte, F., Ratti, C., Guo, D., & Liu, Y. (2023). Comprehensive urban space representation with varying numbers of street-level images. *Computers, Environment and Urban Systems, 106*, Article 102043.

Kelly, C. M., Wilson, J. S., Baker, E. A., Miller, D. K., & Schootman, M. (2013). Using google street view to audit the built environment: Inter-rater reliability results. *Annals of Behavioral Medicine, 45*(S1), 108–112.

Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). Imagenet classification with deep convolutional neural networks. In *Proceedings of the 25th international conference on neural information processing systems - volume 1, NIPS'12* (pp. 1097–1105). Red Hook, NY: USA. Curran Associates Inc.

Li, X., Zhang, C., Li, W., Ricard, R., Meng, Q., & Zhang, W. (2015). Assessing street-level urban greenery using google street view and a modified green view index. *Urban Forestry & Urban Greening, 14*(3), 675–685.

Li, Y., Huang, Y., Mai, G., & Zhang, F. (2025). *Learning street view representations with spatiotemporal contrast*.

Liang, X., Zhao, T., & Biljecki, F. (2023). Revealing spatio-temporal evolution of urban visual environments with street view imagery. *Landscape and Urban Planning, 237*, Article 104802.

Liu, Z., Yang, A., Gao, M., Jiang, H., Kang, Y., Zhang, F., & Fei, T. (2019). Towards feasibility of photovoltaic road for urban traffic-solar energy estimation using street view image. *Journal of Cleaner Production, 228*, 303–318.
Long, Y., & Liu, L. (2017). How green are the streets? An analysis for central areas of chinese cities using tencent street view. *PLoS One, 12*(2), Article e0171110.
Lowe, D. G. (1999). Object recognition from local scale-invariant features. In *, 2. Computer vision, 1999. The proceedings of the seventh IEEE international conference on* (pp. 1150–1157). IEEE.
Luo, Y., Leong, C.-T., Jiao, S., Chung, F.-L., Li, W., & Liu, G. (2023). Geo-tile2vec: A multi-modal and multi-stage embedding framework for urban analytics. *ACM Transactions on Spatial Algorithms and Systems, 9*(2), 10:1–10:25.
Mai, G., Xuan, Y., Zuo, W., He, Y., Song, J., Ermon, S., Janowicz, K., & Lao, N. (2023). Sphere2vec: A general-purpose location representation learning over a spherical surface for large-scale geospatial predictions. *ISPRS Journal of Photogrammetry and Remote Sensing, 202*, 439–462.
Mikolov, T., Chen, K., Corrado, G., & Dean, J. (2013). *Efficient estimation of word representations in vector space*.
Miller, J. (1910). *The building of the city beautiful. 6*. A. Brandt.
Naik, N., Kominers, S. D., Raskar, R., Glaeser, E. L., & Hidalgo, C. A. (2017). Computer vision uncovers predictors of physical urban change. *Proceedings of the National Academy of Sciences, 114*(29), 7571–7576.
Ranftl, R., Bochkovskiy, A., & Koltun, V. (2021). *Vision transformers for dense prediction*.
Rapinel, S., Clément, B., Magnanon, S., Sellin, V., & Hubert-Moy, L. (2014). Identification and mapping of natural vegetation on a coastal site using a worldview-2 satellite image. *Journal of Environmental Management, 144*, 236–246.
Ros, G., Sellart, L., Materzynska, J., Vazquez, D., & Lopez, A. M. (2016). The synthia dataset: A large collection of synthetic images for semantic segmentation of urban scenes. In *2016 IEEE conference on computer vision and pattern recognition (CVPR)* (pp. 3234–3243).
Rundle, A. G., Bader, M. D., Richards, C. A., Neckerman, K. M., & Teitler, J. O. (2011). Using google street view to audit neighborhood environments. *American Journal of Preventive Medicine, 40*(1), 94–100.
Sun, M., Zhang, F., Duarte, F., & Ratti, C. (2022). Understanding architecture age and style through deep learning. *Cities, 103787*.
Van Cauwenberg, J., Van Holle, V., Simons, D., Deridder, R., Clarys, P., Goubert, L., … Deforche, B. (2012). Environmental factors influencing older adults' walking for transportation: A study using walk-along interviews. *International Journal of Behavioral Nutrition and Physical Activity, 9*(1), 85.
Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., … Polosukhin, I. (2017). *Attention is all you need*.
Wang, M., Zhang, X., Niu, X., Wang, F., & Zhang, X. (2019). Scene classification of high-resolution remotely sensed image based on resnet. *Journal of Geovisualization and Spatial Analysis, 3*(2), 16.
Wang, Y., Huang, H., Rudin, C., & Shaposhnik, Y. (2021). Understanding how dimension reduction tools work: An empirical approach to deciphering t-sne, umap, trimap, and pacmap for data visualization. *Journal of Machine Learning Research, 22*(201), 1–73.
Wohlwill, J. F. (1976). Environmental aesthetics: The environment as a source of affect. In I. Altman, & J. F. Wohlwill (Eds.), *Human behavior and environment* (pp. 37–86). US, Boston, MA: Springer.
WorldPop. (2018). *Global 100m covariates*.
Yan, B., Janowicz, K., Mai, G., & Gao, S. (2017). From itdl to place2vec: Reasoning about place type similarity and relatedness by learning embeddings from augmented spatial contexts. In *Proceedings of the 25th ACM SIGSPATIAL international conference on advances in geographic information systems, SIGSPATIAL '17* (pp. 1–10). New York, NY, USA: Association for Computing Machinery.
Yao, Y., Qia, Z., Zijin, G., Weiming, H., Yatao, Z., Xiaoqin, Y., Anning, D., Zhangwei, J., Hong, L., & Guan, Q. (2023). Unsupervised land-use change detection using multi-temporal poi embedding. *International Journal of Geographical Information Science, 37* (11), 2392–2415.
Yeh, C., Perez, A., Driscoll, A., Azzari, G., Tang, Z., Lobell, D., Ermon, S., & Burke, M. (2020). Using publicly available satellite imagery and deep learning to understand economic well-being in africa. *Nature Communications, 11*(1), 2583.
Zhang, F., Salazar-Miranda, A., Duarte, F., Vale, L., Hack, G., Chen, M., Liu, Y., Batty, M., & Ratti, C. (2024). Urban visual intelligence: Studying cities with artificial intelligence and street-level imagery. *Annals of the American Association of Geographers, 114*(5), 876–897.
Zhang, T., Huang, X., Wen, D., & Li, J. (2017). Urban building density estimation from high-resolution imagery using multiple features and support vector regression. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 10*(7), 3265–3280.
Zhang, X., Du, S., & Wang, Q. (2018). Integrating bottom-up classification and top-down feedback for improving urban land-cover and functional-zone mapping. *Remote Sensing of Environment, 212*, 231–248.
Zhou, B., Lapedriza, A., Khosla, A., Oliva, A., & Torralba, A. (2018). Places: A 10 million image database for scene recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence, 40*(6), 1452–1464.
Zhou, B., Zhao, H., Puig, X., Fidler, S., Barriuso, A., & Torralba, A. (2017). Scene parsing through ade20k dataset. In *Proceedings of the IEEE conference on computer vision and pattern recognition*.
Zhou, B., Zhao, H., Puig, X., Xiao, T., Fidler, S., Barriuso, A., & Torralba, A. (2019). Semantic understanding of scenes through the ade20k dataset. *International Journal of Computer Vision, 127*(3), 302–321.